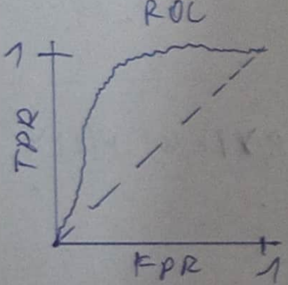
1] Precision $= \frac{TP}{TP+FP}$; Recall $= \frac{TP}{TP+FN}$

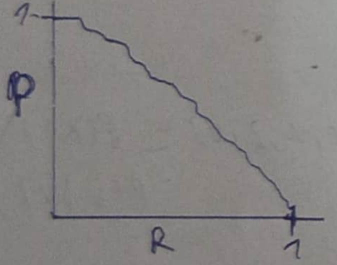
F1-score $= 2\,\frac{P.R}{P+R}$

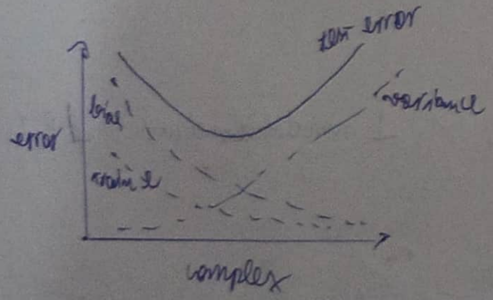
	Macro	Weight	Micro
P	$\frac{1}{k}\sum_{k}^{k} P_k$	$\sum_{k}^{k} w_k P_k$	$\frac{\Sigma\ TP_k}{\Sigma (TP_k + F_{\dots k})}$
R	...	...	
F1	$\frac{1}{k}\sum_{k}^{k} F1_k$	$\sum_{k}^{k} w_k . F1_k$	$\frac{2\,\text{micro}P\,.\,MR}{MP+MR}$

ROC curve: $TPR = \frac{TP}{TP+FN}$; $FPR = \frac{FP}{FP+TN}$

$AUC = \int_0^1 TPR(FPR)\, d(FPR)$







2] DecisionTree

$H(S) = -\sum_{i=1}^{k} p_i \log_2 p_i$

$IG(S,A) = H(S) - \sum_{v \in A} \frac{|S_v|}{|S|} H(S_v)$

$Gini(S) = 1 - \sum_{i=1}^{k} p_i^2$

Impurity split $= \sum_{j=1}^{k} \frac{N_j}{N} \times Gini(S_j)$

Gini gain ($\Delta Gini$) = Gini(parent) − Imp(split)

Pre pruning: set condition during plant grow.
- max_depth ; min_samples_split
  + high speed, save memory
  − easy to get underfitting

Post pruning: Prune after grow.
  + unaffected by underfitting
  − need more resource for training, have to use a part of validation set to ~~tr~~ find tree

$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$

$\tanh'(z) = 1 - \tanh^2(z)$

$\begin{cases} SiLU = z\sigma(z) \\ SiLU' = \sigma(z) + z\sigma(z)(1-\sigma(z)) \end{cases}$

3] MLP & Perceptron

$\left(\frac{1}{u}\right)' = -\frac{u'}{u^2}$; $(\sqrt{u})' = \frac{u'}{2\sqrt{u}}$; $(\sin u)' = u'\cos u$;

$\cos u' = -u'\sin u$; $\tan u' = \frac{u'}{\cos^2 u}$; $\log_a u' = \frac{u'}{u \ln a}$

$(\cot u)' = -\frac{u'}{\sin^2 u}$; $e^{u'} = u' e^u$; $\ln u' = \frac{u'}{u}$

Softmax $\hat{p}_k = \frac{\exp(w_k^T h^{(L)} + b_k)}{\sum_j \exp(w_j^T h^{(L)} + b_j)}$

$\frac{\partial s_m}{\partial w} = s_i(\delta_{ij} - s_j)$ $\delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$

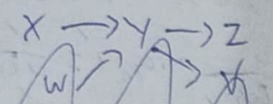
Sigmoid $\frac{1}{1+e^{-z}}$ $\sigma'(z) = \sigma(z).(1-\sigma(z))$

$\sigma(z)' =$ let $u = 1+e^{-z} \Rightarrow f(u) = \frac{1}{u}$

$\frac{\partial \sigma}{\partial z} = \frac{\partial \sigma}{\partial u} \cdot \frac{\partial u}{\partial z} = -\frac{1}{u^2} \cdot (-1)e^{-z}$

$= \frac{e^{-z}}{(1+e^{-z})^2} =$ ~~$\frac{e^{-z}+e^{-2z}}{(1+e^{-z})^3}$~~ $\sigma(z).(1-\sigma(z))$

4] Bayesian ~~networ~~k learning



$P(X,Y,Z,w,V) \propto P(w)\cancel{P(X)}\cdot P(Y|W,X)\cdot P(Z|Y)\cdot P(V|X)$

$$P(\theta|D) = \frac{P(D|\theta)\cdot P(\theta)}{P(D)}$$

$$P(\theta|D) \propto P(D|\theta)\cdot P(\theta)$$

$$MLE = \arg\max_{\theta} P(D|\theta)$$

$$MAP = \arg\max_{\theta} P(\theta|D) = \arg\max_{\theta} P(D|\theta)\cdot P(\theta)$$

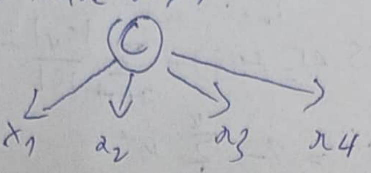
5] Generic

$$P_r(h_i) = \frac{F(h_i)}{\sum_{h\in P} F(h)}$$

- Single point : ~~swap~~ 1 point rotuad
- Two point : 2 point
- Uniform
- Variable-length bit string

6] Bayes network. & Naive Bayes

* Naive Bayes:



$x_i \perp x_{-i} | C$

in case probability distribution continuous:

$$p(x_j; \mu_{ch,d}; \sigma^2_{ch,d}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right)$$

$$\mu = \frac{1}{N_k}\sum_{i=1}^{N_k} x_d^i$$

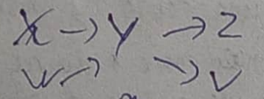
$$\sigma^2_{ch,d} = \frac{1}{N_k-1}\sum_{i}^{N_k}(x_d + \mu_{ch,d})^2$$

Descendant : hậu duệ

fraund : gian lận

⊥ : độc lập ~~(độ~~

* Bayes network



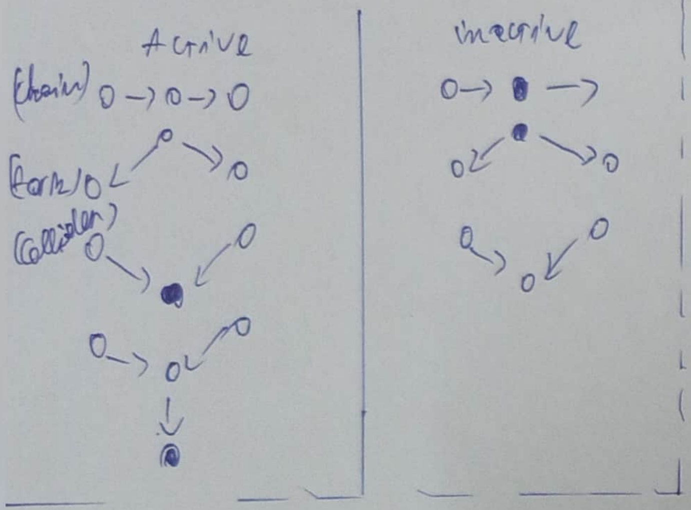
$P(X,Y,w,Z,V) = P(X)P(W)P(Y|X,W),P(Z|Y)$
$P(V|Y)$

$$P(x_1,x_2,\dots,x_n) = \prod_{i=1}^{n} P(x_i|Pa(X_i))$$

$X_i \perp$ Non descendants $(X_i) | Pa(X_i)$

$P(x_1,\dots,x_n) = \prod_{i=1}^{n} P(x_i|Pa(x_i))$ for each $X_i \perp$ nondescendant$(X_i) | Pa(X_i)$

d-separation.


Active
inactive
(chain)
(fork)
(collider)